Midterm Project: UR3e Cobot Laboratory Operations

Participating Team Members: Eva, Andre, Muhammad

Course: ITAI 2374 – Spring 2026

Equipment: UR3e Cobot, Hand-E Gripper, Dorner 2200 Series Conveyor

1. **Project Activities**
   Our team focused on the foundational and advanced deployment of the **UR3e Cobot** in **Robotics Lab N138.** Over two primary lab sessions, we transitioned from basic hardware setup to autonomous, sensor-based programming:

   - **Phase 1: Hardware & Configuration (April 14):** We performed a full system initialization, including powering the robot and navigating the PolyScope GUI on the Teach Pendant. We successfully mounted the Hand-E gripper and performed critical calibrations: **Tool Center Point (TCP)**, **Orientation configuration**, and **Payload/Center of Gravity** definitions to ensure accurate robotic movement.

   - **Phase 2: Motion & Basic Programming:** We practiced distinct motion types (**MoveJ** and **MoveL**) and learned to use the Align tool and Home position features. We transitioned into programming by setting waypoints, renaming them, and establishing a “Before start-sequence” to manage safety.

   - **Phase 3: Advanced Automation (April 21):** We developed a sophisticated pick-and-place loop. This involved expanding the program tree to include seven named waypoints and implementing “If object detected” conditional logic. This advanced phase allowed the robot to interact intelligently with the conveyor system rather than just following a static path.

2. **Challenges and Obstacles**
   While the UR Academy modules provided a strong theoretical foundation, transitioning to the physical UR3e hardware presented several technical hurdles that required active troubleshooting:

   - **TCP Orientation and Precision:** During the tool configuration phase on April 14, our team encountered difficulty with the **Orientation configuration** necessary for the robot to understand its tool's position in 3D space. Initially, the robot's manual TCP inputs were slightly off, leading to subtle misalignments. We resolved this by meticulously revisiting the **"Configuration of a Tool Using the Wizard"** tutorial and performing micro-adjustments until the Hand-E gripper maintained perfect perpendicularity relative to the conveyor belt.

- **Navigating Administrative Permissions:** A recurring obstacle was the restrictive nature of the PolyScope software's default user settings, which prevented us from accessing advanced programming features. We overcame this by identifying and applying the **Operational mode password**, which granted us the administrative access necessary to define custom waypoints and program folders.

- **Payload and Inertia Management:** On April 21, as we increased the complexity of our pick-and-place loop, the robot began triggering protective stops. We realized that failing to correctly set the Payload and Center of Gravity in the early stages caused the robot's motion to become jerky. By recalculating the payload for the Hand-E gripper and updating the software parameters, we successfully smoothed the motion and ensured the system operated within its safe momentum limits.

- **Sensor Logic and Gripper Sensitivity:** On April 21, we encountered a challenge where the Hand-E gripper sensor triggered prematurely or became "stuck" due to detected resistance. This taught us that sensor thresholds must be precisely calibrated to distinguish between the work surface and the target payload.

  Video: Sensor Sensitivity Troubleshooting (April 21)

  https://youtu.be/E3oak1pb_vA?si=vqz56tSsP-RJQqzV

  This footage demonstrates a real-world sensor conflict where the gripper's force detection logic halted the program. This event was a critical turning point for the team, requiring us to recalibrate the gripper's "Object Detection" parameters to ensure continuous operation.

3. **Team Accomplishments**

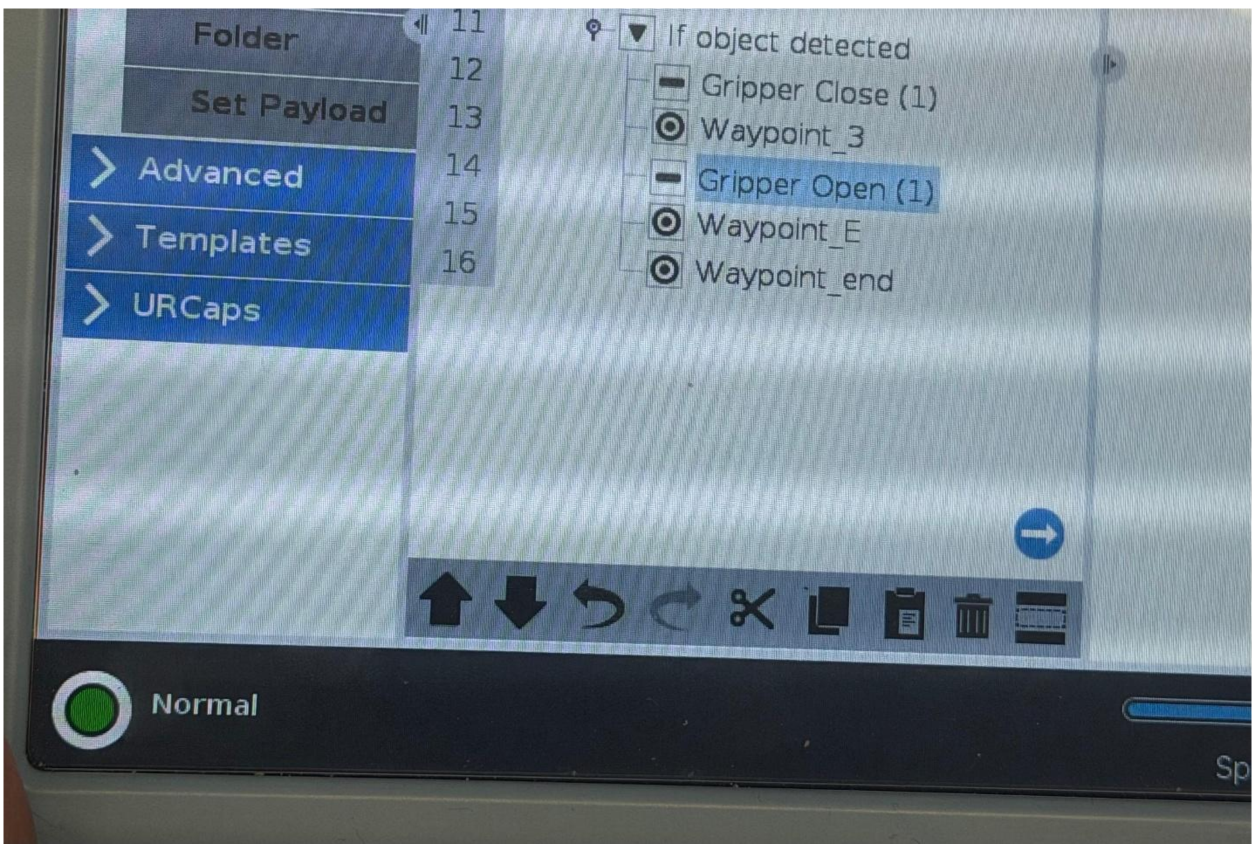
Our team successfully transitioned from E-learning simulations to a functional physical setup in **Lab N138.**

- **Integrated Setup:** We successfully integrated the **UR3e** with a **Hand-E Gripper** and aligned it with the **Dorner 2200 Series conveyor.**

- **Autonomous Logic:** We created a program that starts at a **“Home”** position, moves to a **“Hover”** waypoint, and executes a controlled descent based on object detection.

- **Results:** The robot maintained perfect perpendicular alignment throughout the motion, confirming that our TCP and orientation settings were correct.

Figure 1: Initial Waypoint Setup (April 14)

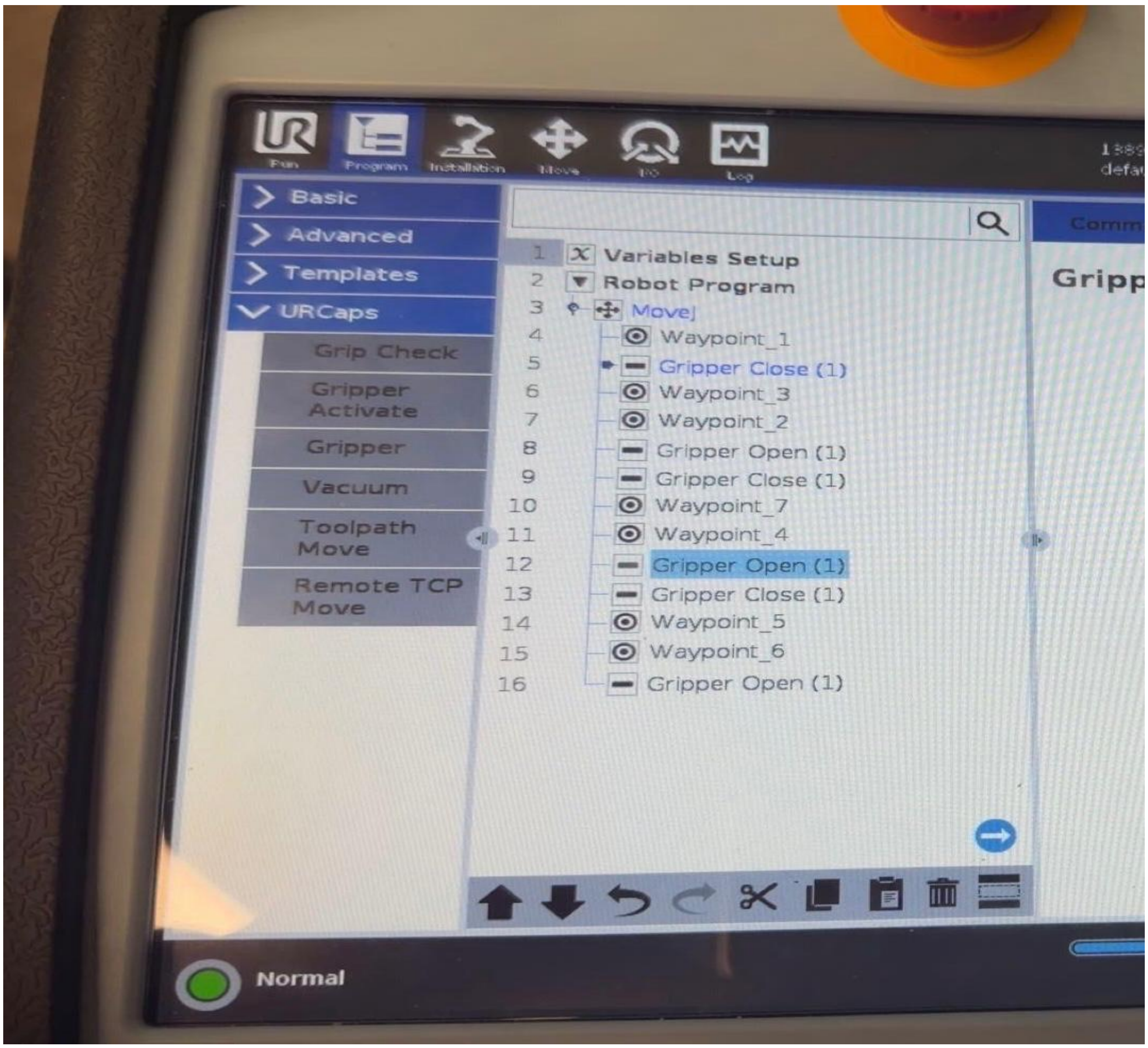


Video 1: Programmed Waypoint Sequence (April 14). Click the link below to watch the video:

https://youtube.com/shorts/oE1sZEHLwzY?si=-hUoOL8lHpCqD9ed

Video 1 shows a 10-second demonstration of the foundational waypoint sequence and smooth vertical motion established in our first session.

Figure 2: Advanced Logic Tree (April 21)



Video 2: Full Automated Cycle (April 21)

https://youtu.be/w04tgsb4hGg?si=uFBYqxlELW4V0JxZ

Video 2 shows the robot successfully executing the complete automated loop, transitioning smoothly between approach, pick, and place waypoints.

Video 3: Dynamic Pick-and-Place with Active Conveyor (April 21)

https://youtu.be/zDXE_B3pTqA?si=6NM4RYQD8znWy6l6

Video 3 showcases the UR3e successfully identifying and picking targets while the conveyor is in motion, validating the timing and sensor logic synchronization.

4. Learning Outcomes

Beyond the basic operation of the UR3e, this project provided our team with several high-level takeaways regarding industrial automation:

- **The Synergy of Software and Physics:** We developed a deep understanding of how software parameters directly dictate physical behavior. Specifically, the relationship between **Payload/Center of Gravity** and the robot's **momentum sensors**. We learned that "correct" settings are not just for accuracy but are essential for preventing mechanical wear and nuisance "Protective Stops."

- **Workflow Optimization:** We discovered that a "hybrid" programming approach is the most efficient. By using **Free Drive** to physically guide the arm into approximate positions, we saved significant time. However, we also learned that industrial precision requires following up with the **Teach Pendant's** manual coordinate entry to ensure waypoints are mathematically aligned for repeatability.

- **Logic-Based Problem Solving:** Moving from a linear "Move-to-Point" program to a conditional **"If-Then" Logic loop** shifted our perspective from simple robotics to systems integration. We learned that a successful program must account for the presence (or absence) of a workpiece to truly be considered "automation."

5. Critical Analysis

In evaluating the **UR3e equipment** and **Universal Robots Academy** curriculum, our team reached the following conclusions regarding their educational value:

- **Hardware Accessibility:** The UR3e's PolyScope interface is exceptionally intuitive for students. The ability to see the program tree update in real-time as we adjusted waypoints allowed for immediate feedback and rapid prototyping of our pick-and-place sequence.

- **Digital Twin Effectiveness:** The UR Academy simulations were instrumental. Despite the instructional videos utilizing a **UR20 model**, the high degree of software parity across the UR ecosystem meant that our transition to the physical **UR3e** in Lab N138 was seamless. The logic, icon sets, and safety configurations were identical, proving the effectiveness of "simulation-to-reality" training.

- **Conclusion:** The combination of the **Dorner conveyor, Hand-E gripper,** and **UR3e cobot** provided a perfect "micro-factory" environment. This setup was highly appropriate for this course level, as it allowed us to encounter—and solve—the same synchronization and sensor-calibration issues faced by professional automation engineers.